Operations Research for Health Care

Lesson 9/15

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UFMG, Belo Horizonte, Brasil
18.12.2024

Outline of this lecture

Primary care facility location: Case 1

Primary care facility location: Case 2

Primary care facility location: Case 3

Primary care facility location: Cases 4 and 5

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An hierarchical location-allocation model for PHC [Hodgson, 1988]

Location-allocation models can make PHC facilities more accessible.

- **Traditional models** have failed because health care systems are hierarchical in nature and benefits and utilisation decline with distance.
- The **Contribution**: A location-allocation model where benefits increase with facility level and decreases exponentially with distance.
- **Results**: traditional P-median *can be worse than a simple strategy* of locating facilities from the highest to the lowest level in centres of decreasing population.

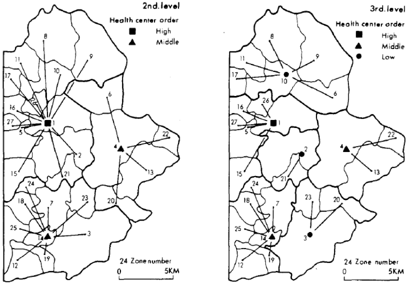

Fig. 4. The *P*-MEDIAN result, $\alpha = 0.0$, $\beta = 0.0495$.

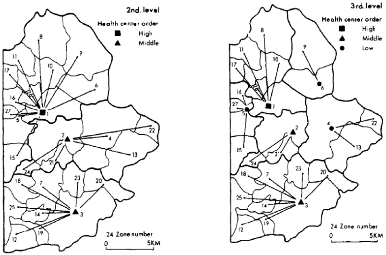

Fig. 5. The SIZE-DOWN result, $\alpha = 0.9260$, $\beta = 1.0$.

An hierarchical location-allocation model for PHC [Hodgson, 1988]

Intro: Imbalance between the provision of urban and rural facilities. In its single-level form, the P-median model is by far *the most widely used location-allocation approach*.

- The assumptions upon which the P-median model are based are unrealistic: *all patients do not attend the closest facility*; therefore, minimising the distance from the nearest facility is not an appropriate goal.
- The higher-order facility *may* have more staff, and *a lower expected waiting time*.
- The patient *may* feel that the higher-order facility can perform the low-level service better (that a doctor used to dealing with serious problems will band the finger better).
- A trip to the facility *may be combined* with a shopping trip, for which the larger centre may be more attractive.
- **Service quality** is the second most important determinant of distance decay. Higher-order facilities are more successful in attracting longer distances patients.

An hierarchical location-allocation model for PHC [Hodgson, 1988]

Proposal:

- **A model** considering differential attractiveness based on facility level.
- **Attractiveness** factor with a negative exponential distance decay function.
- **ACTUAL** (Service sizes, Use): 1 High-level (6, 0.1), 2 Middle -Level (3, 0.3), and 3 Low-level (1, 0.6)
- **P-MEDIAN**: Classic demand weighted distance model.
- **SIZE-DOWN**: Simple allocation from highest to lowest level in centers of decreasing population.

An hierarchical location-allocation model for PHC [Hodgson, 1988]

I : Aggregate demand area.

J : Potential facility locations

K : Facility level

- S_j^α: Facility level of service at j. Service levels (sizes) are estimated at 6, 3 and 1.
- U_k: Relative usage of service k. Proportional to the number of places offering (from high to low) service k, i.e. 0.1, 0.3 and 0.6.
- α: Effect of attractiveness due to size.
- β: Effect of distance on attractiveness.
- P_i: Demand for origin i.

$$(S_j^\alpha)^{-\beta d_{ij}} \tag{1}$$

The hierarchical model treats facilities successively-inclusive hierarchy, and maximizes benefits as:

$$max. \sum_{k \in K} U_k \sum_{i \in I} P_i * max_{j \in J} (S_j^\alpha)^{-\beta d_{ij}} \tag{2}$$

An hierarchical location-allocation model for PHC [Hodgson, 1988]

- The results vary considerably with variation in α and β, however.
- This depends on data, that are difficult to obtain.
- The case-study was evaluated with 27 urban and rural agglomeration served as focal point for demand-zones.
- **Calibration** of the attendance model with real world interaction data is essential.

Outline of this lecture

Innovative Applications of O.R.: Location and reorganization problems: The Calabrian health care system case [Guerriero et al., 2016]

- Italian health care system: *more efficiency* while *containing costs*.
- *Compare* the existing health care service network with classic location problem.
- Then, optimize considering national and regional constraints.
- Extensive computational study and real data.

Reorganization: number or the position of facilities, to their capacities and/or types of supplied services. Varible most discussed is demand.

Innovative Applications of O.R.: Location and reorganization problems: The Calabrian health care system case [Guerriero et al., 2016]

In this paper:

First : describe the current localization of public hospitals (performance index);
Second : Compare with classical optimization facility location models;

Scenarios:

1. **Minimize number of public hospitals to be opened** without taking demand, *then* apply a covering model to evaluate quality of service solution (% demand met);
2. Satisfy demand by reallocating part of the demand to hospital whose capacity can be expanded within some limit. Index: ALOS, capacity as departments, utilization of beds.

Results: **Current** hospital network setting leave some demand *uncovered*. Scenarios:

- **[1]** A network with unacceptable high percentage of unmet demand.
- **[2]** Complete demand satisfaction and enable alternative solutions.

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PHC Center: Location-Allocation with Multi-Objective Modelling

[Miç et al., 2019]

- Due to war, people in Siria face troubles with regard to health, shelter, food and other needs.
- Focus identifying the Primary Health Care Center (PHCC) locations. Data is a sample of 23 sub-districts with 388 communities.
- Geographical Information System (GIS - Maps) with optimization package.
- Initial solution: 60 PHCCs out of 77 candidate PHCCs, but the model located 42 PHCCs in total allocating 380k people, cost of US$ 1M, variable cost U$163k.
- These humanitarian emergencies affect people's basic needs (food, clothing, shelter, water) and security/health.
- In this context, it is worth mentioning **the importance of primary health care centres** (PHCCs) at the *local level*.
- PHCC are the key structural and operational components of public health services.

PHC Center: Location-Allocation with Multi-Objective Modelling

[Miç et al., 2019]

- **Methodology**: The **proposed** multi-objective (*weighted goal programming*) model aims to maximise the number of *people served* and minimise *overall costs* at the same time.

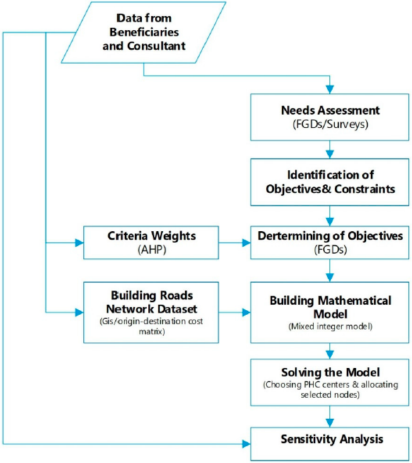


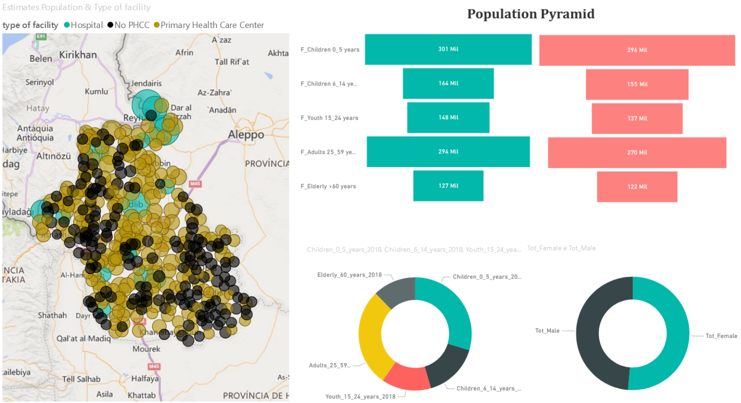
Figure 1. Proposed methodology adopted in the study.

PHC Center: Location-Allocation with Multi-Objective Modelling

[Miç et al., 2019]

Case study and results:

- **Data**: Available at interactive link (https://goo.gl/x2GjVv).



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PHC: Assess performance, re-design with optimisation, and distribute patients for equitable workloads [Elorza et al., 2022, Teymourifar and Trindade, 2024]

- **Proposal 1**: A maximum coverage facility location methodology founded on need/offer/demand quantification to assess existing **PHCC** performance and assist in its re-design [Elorza et al., 2022].
  - This is a *service capacity problem*, since the municipality' PHCCs are well geographically distributed, but the service offer in each PHCC does not match the population needs.
  - Solutions are obtained if (i) *budgetary* or (ii) *minimum attention volume constraints*.
  - Solutions: Offer/need, personnel, material, variable and relocation costs.
- **Proposal 2**: An approach to evenly distributing healthcare workloads, aimming to a more **equitable allocation of tasks** using optimisation to dispatching rules (multi-objective model grounded in the concept of sectorization) *to balance patient distribution* and increase healthcare accessibility. Results are simulated [Teymourifar and Trindade, 2024].
  - See: Acknowledgments, Data availability statement (Github, Youtube), Supplemental data, and Funding.

References

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